Learn link

GUIDE

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Problem Statement

- Challenges
  - Finding a platform that integrates well and helps teams work together.
  - Existing solutions don't effectively facilitate communication, knowledge sharing, and meaningful discussions.
- Missing Features
  - Seamless communication tools.
  - AI-driven insights for informed decision-making.
  - Intuitive design for easy use.
- Consequences
  - Users struggle to organize their ideas efficiently.
  - Learning processes are not streamlined.
- Need for a Solution
  - A comprehensive platform that boosts productivity.
  - A platform that fosters academic excellence through enhanced collaboration and resource sharing.

Solution Approach

Real-Time Collaboration and Communication

- Voice calls for discussions, brainstorming, and team learning.
- Voice communication, annotations, and co-editing to enhance teamwork.

Content Sharing and Management

- Upload and share educational materials, question papers, and brainstorming artifacts.
- Save content from brainstorming session

AI-Powered Tools

- Summaries: Create concise summaries of documents.
- Quiz Generation: Make quizzes for self-assessment.
- Focus Area Extraction: Identify key questions and categorize them.

Modules

User Authentication

- User Login
- User Registration
- Google Login
- Account Settings

AI Integration

- AI-powered Document Summarization
- Focus Area
- Quiz Generation
- Flash Cards

Collaboration

- Live Cursor
- Collaborative canvas
- Voice Communication

File Processing

- State Updates
- File Limit Enforcement
- Duplicate Prevention
- File Type Identification

Technology Stack

Front End

React js : React.js is a powerful JavaScript library used for building user interfaces.
Scan/cd : Frontend component Library

Back End

Node js : JavaScript runtime for server-side applications.
Express js : Web application framework for Node.js that simplifies server-side development by providing robust features for building web applications.

API Used

Gemini API : AI-powered text and media processing.
Auth0 : Auth0 enables secure user authentication and authorization

Implementation Details

<u>User Authentication</u>

1.Auth0 Integration:

- Auth0Provider wraps the app for authentication.
- Configured with domain, clientId, and redirect_uri from environment variables.

2. Session Management:

- Sessions persist using localstorage.

3. Redirect Handling:

- Redirects users to /dashboard after login.
- Custom callback ensures proper navigation.

4.Error Handling:

- Logs authentication errors for debugging.

Implementation Details

<u>AI Integration Implementation</u>

1. Purpose:
   - Enhances user experience with intelligent features.

2. Integration:
   - AI services accessed via APIs or libraries

3.Frontend:
   - React components interact with AI for real-time responses.

4.Optimization:
   - Asynchronous calls and caching for performance.

<u>Collaboration</u>

1.Real-Time Features:
   - Enables multiple users to interact or work together in real time (e.g.,Live cursor or Voice calls).

Implementation Details

2.Backend Integration:
- Uses WebSockets, REST APIs, or third-party services for real-time data synchronization.

3.Frontend Interaction:
- React components dynamically update the UI based on collaborative actions.

4.User Roles:
- Implements role-based access control for managing permissions in collaborative environments.

<u>File Processing</u>

1. File Upload:
   - Users can upload files via a React-based UI.
   - Files are sent to the backend using APIs.

Implementation Details

2.Backend Handling:

- Files are processed on the server
- Supports PDFs.

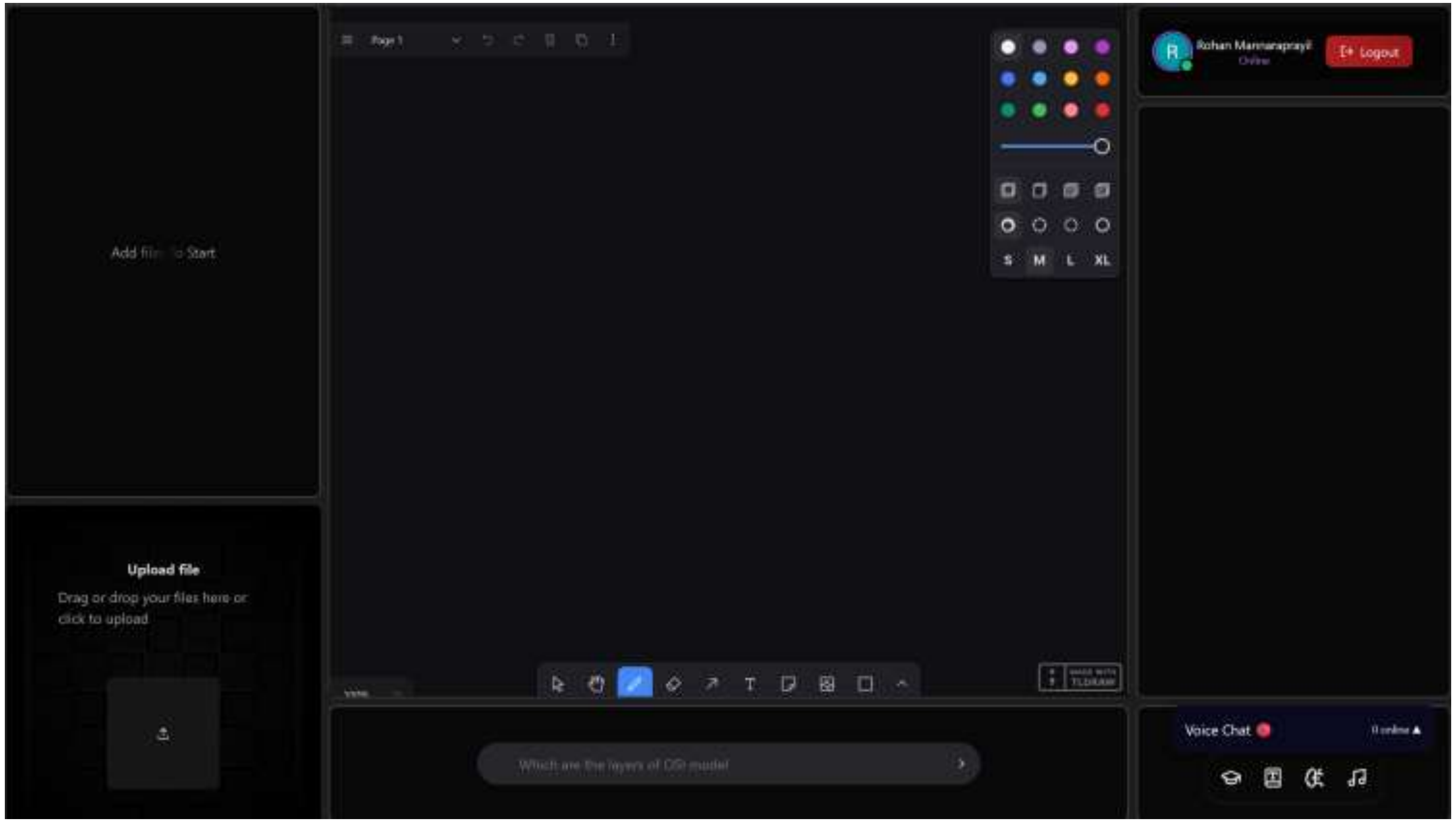
3.Feedback to Users:

- Real-time progress indicators for uploads.
- Error messages for invalid files or processing failures.

4.Security:

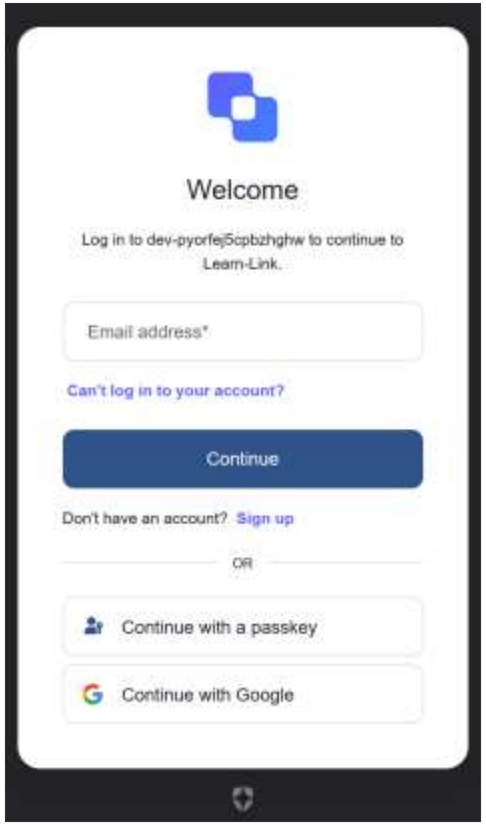
- Implements file size limits and type validation to prevent malicious uploads.

Screenshots



Img1:Dashboard

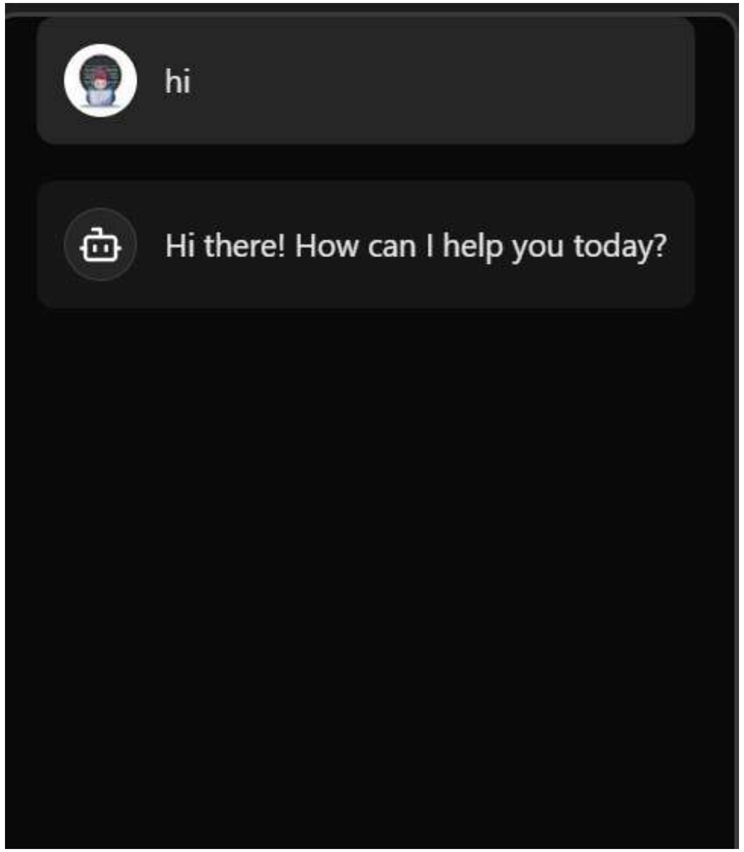
Screenshots



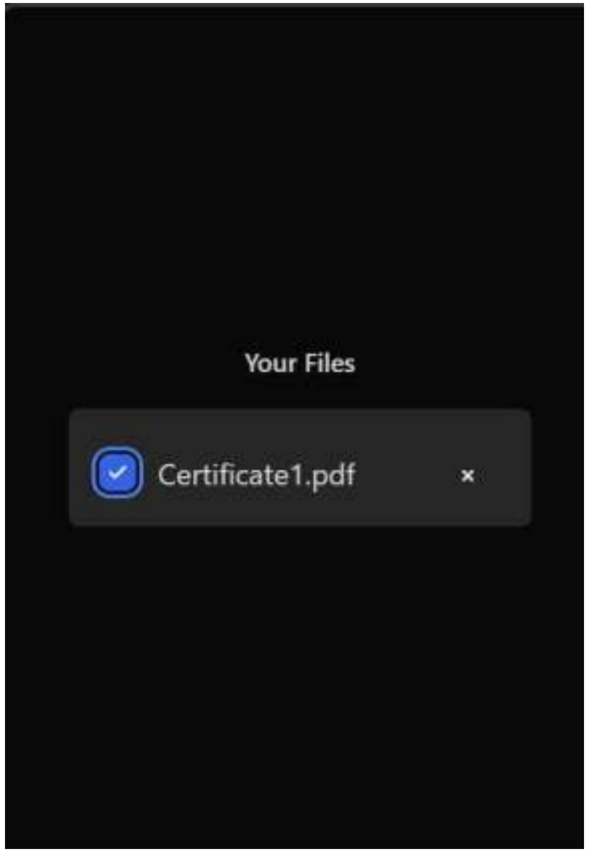
Img2:User Authentication

Img3:Voice Chat

Screenshots



Img4:AI Integration



Img5: File processing

Conclusion

- Learn-Link stands as a testament to the transformative power of technology in education.
- By addressing key limitations of traditional learning such as collaboration, interactivity, and efficiency, Learn-Link has set a new standard for productive and engaging learning experiences.
- Through its innovative use of real-time tools and AI, the desktop website not only provides users with a dynamic and collaborative platform to explore but also offers a glimpse into the future of learning.
- The success of Learn-Link highlights the potential for technology to revolutionize education, offering a wide range of benefits that transcend traditional learning methods.
- From enhanced teamwork and personalized insights to streamlined resources and secure access, this platform opens up new possibilities for learners. As web technologies continue to advance, the potential for innovative and transformative learning experiences will only continue to grow, making Learn-Link a pioneering example of the future of education.

References

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Thank you